ELANGO S

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CAREER OBJECTIVE

Aspiring Machine Learning Engineer seeking an entry-level opportunity to apply academic knowledge and hands-on project experience to solve real-world problems, contribute to impactful AI solutions, and grow as a professional in a collaborative engineering environment.

EDUCATION

B.Tech Artificial Intelligence and Data Science
Nehru Institute of Engineering and Technology, Anna University **2027 | CGPA: 7.9 / 10**

Higher Secondary (12th)
State Board of Tamil Nadu — GHSS **2023 | 83.33%**

Secondary (10th)
State Board of Tamil Nadu — GHSS **2021 | Pass**

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS

AI / Machine Learning: Machine Learning, Deep Learning, Scikit-learn, PyTorch

Data & Visualization: Pandas, NumPy, Matplotlib

Web Development: React, FastAPI, REST APIs, Bootstrap

Tools & Platforms: Git, GitHub, MCP

PROJECTS

AI-Generated Image Detection and Fraud Prevention System

- Developed a deep learning system to verify image authenticity by classifying uploaded images as **Real or AI-Generated**, targeting image-based fraud and digital content verification.
- Implemented an **EfficientNet-B0/PyTorch** classification pipeline with image preprocessing, confidence scoring, and a **Python FastAPI web interface** for instant analysis.

Crop Yield Prediction Web Application

- Built an agricultural prediction system to estimate **crop yield (kg/hectare)** from area, crop type, year, rainfall, pesticide usage, and temperature.
- Integrated a **Random Forest Regressor** with a **Flask REST API** and **React, TypeScript, Tailwind CSS, and Recharts** frontend for prediction, model metrics, and feature-importance visualization.

Customer Segmentation K-Means Dashboard

- Analyzed the Olist Brazilian E-Commerce dataset to identify customer groups based on **Recency, Frequency, Monetary value, purchasing behavior, and payment patterns** for targeted marketing and retention.
- Developed a **K-Means clustering pipeline** with Elbow Method, Silhouette Score, and PCA, exposed through a **Flask REST API and React dashboard** with segment prediction and business recommendations.

Learning Path Generator with Model Context Protocol (MCP)

- Created a personalized learning-path system that converts a user's **learning goal and time constraint** into a structured learning plan using **MCP and Google AI Studio API**.
- Integrated **YouTube, Google Drive/Notion, and Pipedream** with a **Streamlit interface** to provide learning resources, organize content, and track progress.

INTERNSHIP EXPERIENCE

Machine Learning Intern – Nxtlogic Software Solutions **Dec 2025 – Jan 2026**

- Completed **Machine Learning internship training**, gaining practical exposure to model development, data preprocessing, and ML workflows.
- Applied machine learning concepts through structured hands-on training and technical activities.

AI-ML Virtual Internship – AICTE **Jan 2025 – Mar 2025**

- Completed a **10-week AI-ML Virtual Internship** through the AICTE–EduSkills program with an AWS Academy-provided curriculum.
- Built foundational knowledge of **Artificial Intelligence and Machine Learning** concepts through structured technical training.

Data Science Master Virtual Internship – AICTE **Jul 2025 – Sep 2025**

- Completed a **10-week Data Science Master Virtual Internship** supported by Altair.
- Developed practical knowledge of **data analysis, data science workflows, and analytical techniques**.

Python Full Stack Developer Virtual Internship – AICTE **Oct 2025 – Dec 2025**

- Completed a **10-week Python Full Stack Developer Virtual Internship** through AICTE–EduSkills.
- Gained practical exposure to **Python-based full-stack and web application development**.

CERTIFICATIONS

- Data Analytics with Python – NPTEL, IIT Roorkee | Elite
- Introduction to Machine Learning – NPTEL, IIT Madras
- Data Science for Engineers – NPTEL | Elite
- IBM Machine Learning with Python Professional – Coursera
- IBM Generative AI Engineering Professional – Coursera
- Google Data Analytics Professional – Coursera

DECLARATION

I confirm that all details mentioned in this resume are true and complete to the best of my knowledge.

Signature

